

Natural Capital and the Measurement of Productivity*

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Abstract

Current practice defines productivity as output that is not explained by measured economic inputs. Yet, many seemingly important inputs, e.g., natural capital, are omitted from the set of measured economic inputs and may confound measurement of productivity. We investigate how omitting natural capital as an input to production biases estimates of productivity growth. Empirical correlations between benchmark data on productivity growth and environmental change at the national level suggest the omission of natural capital biases these measures. We provide theoretical conditions under which better measurement of omitted environmental inputs reduces estimation error. Calibrated experiments based on data from the World Bank suggest that the inclusion of natural capital on the margin will improve measurement for most countries.

Keywords: Total Factor Productivity; National Accounts; Natural Capital; Measurement

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*All errors are our own.

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Innovation and technological change over the past two centuries drove an unprecedented increase in economic activity. Industrialization simultaneously muddled societal awareness of fundamental connections between humans and nature. Society’s interest in tabulating the stocks of nature and their associated flows waned, and attention shifted to tracking produced and human capital over time. Theodore Roosevelt noted this phenomenon at the outset of the 20th century:

“With the rise of peoples from savagery to civilization, and with the consequent growth in the extent and variety of the needs of the average man, there comes a steadily increasing growth of the amount demanded by this average man from the actual resources of the country. And yet, rather curiously, at the same time that there comes that increase in what the average man demands from the resources, he is apt to grow to lose the sense of his dependence upon nature.” ([Roosevelt](#)).

Formal systems of economic measurement have followed this trend; existing structures provide little guidance for separating nature’s contribution to production from the contributions of human or produced capital. Human labor, vessels, and sonar are commonplace inputs to industrial fishing and measured by national accountants. Equally important is the fish stock and its ability to replenish over time. However, the fish stock, as an input to production, is often excluded from economic measurement. This continues to be the case despite managed fish stocks (those within exclusive economic zones) being a textbook example of a non-produced natural asset scheduled for measurement under modern standards like the System of National Accounts ([European Commission and Development](#)).

This paper examines how the omissions of the environment as a factor in production affect a leading metric of economic progress produced under current national accounting standards, total factor productivity (TFP). TFP is defined as the growth in output unexplained by growth in inputs; it has been a benchmark metric for economy-wide technological progress and innovation over the past century ([Solow](#)). Changes in TFP over time play a crucial role in shaping governments’ budget projections and policy agendas over long time horizons ([Congressional Budget Office](#); [Council of Economic Advisers](#)). Starting with Kuznet’s seminal work report to congress in 1934, the broader development of statistical, economic, and environmental policy, has focused on measuring output and productivity growth by tracking capital and labor income over time ([Kuznets](#)). While governments historically emphasized tracking natural resources for the sake of posterity and avoiding overuse ([Roosevelt](#)), features of the environment beyond land were omitted from the transition to the double entry-based national accounting systems used today.

While the importance of the environment as a factor for economic activity waned in the eyes of governments during the 20th century, it became increasingly salient as a source of waste management and a determinant of human health. The relationship between governments and the environment changed from one of planning for “wise use” to protection and regulation. Governments’ use of environmental data shifted from tracking the productive potential of the local environment to monitoring indicators used for evaluating and enforcing policy. This paradigm shift

in environmental measurement coincided with the development of modern national statistical systems that drove a massive expansion of economic data collection and curation. Despite massive efforts by governments to collect data on environmental indicators associated with human and animal health outcomes, environmental assets (e.g., fresh water or fertile topsoil) remain segregated from economic accounts.¹ While national accounts standards call for the inclusion of many forms of natural capital as non-financial capital ([European Commission and Development](#)), few countries collect and organize these data with the same fidelity as output or factor inputs. When they do, they are seldom used to measure TFP.

Despite the limited progress incorporating natural capital in national accounts, environmental factors have become increasingly salient for private economic agents. Insurers, producers, and households increasingly demand better environmental data ([Yergin](#); [Boomhower et al.](#)). Unlike regulators, these agents track natural capital stocks because changes in the environmental services flowing from these stocks have material consequences for their economic decision-making. We argue here that the persistent omission of environmental assets from national accounting frameworks also impedes government economic decision-making; the changing state of the environment is material for forecasting growth, irrespective of environmental or sustainability goals. Failing to account for how the environment determines economic growth may cause governments to overestimate the ability for society to substitute away from natural resources and misattribute the effects of economic policy ([Nordhaus and Tobin](#); [Brandt, Schreyer, and Zipperer](#)). The results we present below suggest the potential magnitude of gains in measurement accuracy are substantial. Across all countries in our sample, the median level of improvement (measured by reduction in mean squared error for estimated average TFP growth) is 10 basis points, or a little over 10 percent of median average TFP growth at the country level over that same period. For the case of the United States, when this level of error reduction is compounded over a standard (10 year) budget horizon it is four to five times the annual year-on-year forecast error of the Congressional Budget Office’s deficit projections, and corresponds to revenues larger than the estimated aggregate cost of compliance with the Clean Air Act ([Congressional Budget Office](#); [U.S. Environmental Protection Agency](#)).

Existing Literature: Tobin and Nordhaus motivated the inclusion of natural capital when accounting for growth by appealing to its amenity value (towards their broader measure of economic welfare) as well as the productive value of land ([Nordhaus and Tobin](#)). Subsequent work incorporating the environment in the development accounting literature has primarily focused on improving measures of the share of national income attributable to reproducible capital (e.g., [Mankiw, Romer, and Weil](#); [Caselli](#); [Caselli and Feyrer](#); [Monge-Naranjo, Sánchez, and Santaaulalia-Llopis](#)). These studies apportion capital income between returns on physical investment and resource rents in order to correct for the effects of the latter when comparing the marginal product of produced capital and its share of national income across countries ([Caselli and Feyrer](#); [Monge-Naranjo, Sánchez, and Santaaulalia-Llopis](#)). Broadly, the literature has highlighted the importance of correcting for

¹For example, it was only in the early 2020s that the United States began developing economic accounts for land ([Wentland et al.](#)).

resource rents when calculating the returns to capital across countries while noting that the amount of residual output unexplained by factor inputs remains high (Hsieh and Klenow). We add to this literature in characterizing explicitly how introducing previously-omitted factors of production has ambiguous effects on the bias in measurement of output elasticities when the additions are marginal and do not render the growth-accounting equation correctly specified.

A nascent strain of literature focuses explicitly on how including natural capital affects growth accounting-based estimates of productivity. Our contribution is closest to that of Brandt, Schreyer, and Zipperer, who were the first to our knowledge to study the effects of adjusting existing TFP estimates at the national level to account for the extraction of subsurface minerals and fossil fuel. The authors show these adjustments depend on the relative rate of growth in the use of non-renewable resources to that of changes in other inputs, which leads measurement of natural capital to have ambiguous effects on growth estimates. Bastien-Olvera and Moore perform a similar exercise when calibrating a regional integrated assessment model used for adjusting the social cost of carbon to account for non-market ecosystem services. Finally, Freeman, Inklaar, and Diewert highlight the importance of accounting for the fact that not all countries extract the same set of non-renewable resources domestically when comparing productivity levels across countries. This is distinct from our exercise comparing productivity growth within countries across time, but presents a methodology that would allow our calibration exercise to be applied to more granular constructions of TFP (e.g., those deployed in the Penn World Tables Feenstra, Inklaar, and Timmer). We add to this literature by explicitly characterizing under what conditions the measurement of non-renewable resources will reduce errors in estimated TFP, as opposed to calculating the magnitude of the adjustments induced by inclusion. One reason to focus on error reduction is that the approach and intuition likely extends to other omitted capital stocks, such as renewable natural capital.

The paper proceeds in three steps. Section 1 tests whether existing, widely used metrics of TFP suffer from omitted variable bias due to the omission of natural capital stocks beyond those with readily available market prices. We show *prima facie* evidence that a modest subset of natural capital stocks are strongly correlated with benchmark measures of international productivity from the academic literature and national statistical agencies. Section 2 derives sharp conditions under which including a partial accounting of natural capital improves measurement of TFP. Even with perfect measurement of all inputs in production, the inclusion of previously-omitted natural capital stocks has ambiguous effects on the bias and error of estimates of productivity growth. Finally, Section 3 uses data on natural resources from the World Bank’s Changing Wealth of Nations (CWON, World Bank) to calibrate an empirical analog of the theoretical setting in Section 2. Using estimated growth in indices measuring the aggregate use of non-renewable and renewable natural capital at the country level between 1996 and 2019, we show that for plausible parameters including non-renewable capital when estimating TFP reduces in-sample error across all but three of the sample of 100 countries with available data.

1 Residual-Based Productivity Measurement

Neoclassical economics models output being attributable to two sources: factors of production (factor inputs) and technology (Solow). We consider measuring productivity for a version of this economy with four factor inputs: produced capital, labor, non-renewable natural capital, and renewable natural capital, denoted by the vector $\mathbf{X}(t) \triangleq \{K(t), L(t), N_1(t), N_2(t)\}$ where output is governed by a production function, $F(\mathbf{X}(t))$, taking the four inputs as arguments. There are two interrelated definitions of productivity for this economy. The first defines total factor productivity (TFP) as a summary statistic for the level technology available to society at a given point in time. Final output is defined as product of this summary statistic $TFP(t)$ and inputs to the production function at each instant:

$$Y(t) \triangleq TFP(t) \times F(\mathbf{X}(t)). \quad (1)$$

Total factor productivity is a theoretical modeling device; an index number for how technology determines the instantaneous possibility production frontier each instant.² It is a parsimonious treatment for how forces such as innovation and creative destruction can shift what society can make given the available inputs (Romer; Aghion and Howitt). Under standard assumptions for the functional form of the production function $F(\mathbf{X}(t))$ (e.g., the Cobb-Douglas or transcendental logarithmic forms), growth in output can be expressed in a log-linear form

$$g_Y(t) = g_{TFP}(t) + \sum_{x \in \mathbf{X}(t)} \gamma_x g_x(t), \quad (2)$$

where the $g_x(t)$ terms are the instantaneous growth rates and γ_x are the output elasticities for each input.³ Equation (2) is known as the *growth accounting equation*; it decomposes output growth between factors of production and technological change.

The definition above is a bottom-up approach; productivity and factor inputs are the economic primitives that explain the entirety of growth in output (Solow). The second definition of productivity growth is grounded in measurement and is defined from the top-down. When output and factor inputs are the primitives of the model, TFP growth is instead defined as

$$g_{TFP}(t) \triangleq g_Y(t) - \sum_{x \in \mathbf{X}(t)} \gamma_x g_x(t). \quad (3)$$

TFP growth is the portion of growth in output not explained by growth in inputs (Comin). These two definitions are easily conflated because of the mathematical equivalence that occurs when they are applied to a model of the economy. However, from a conceptual basis, they are distinct objects:

²We assume throughout this paper that technical change is Hicks neutral (i.e., follows 1). A more general formulation of technical change from a modeling standpoint is the partial derivative of the function determining aggregate output in time (Solow).

³Output elasticities need not be fixed over time (e.g., if production takes a constant elasticity of substitution form) but we adopt that assumption here.

the first assumes productivity is a primitive determinant of economic output independent of the level of inputs, while the other defines it as the residual portion of output unexplained by inputs. Both objects are theoretical; neither an indicator for the level of technology in the economy or the production function governing the economy are observable to an econometrician. The top-down definition of productivity will be the focus of this paper, as it is the relevant one for measurement

In practice, TFP growth is typically measured as the portion of output growth not explained by growth in a set of *observed* factor inputs, which we group together as the set $\mathbf{I}$. Factor inputs (e.g., produced capital and labor) and production of final output are measured by national statistical agencies at regular frequencies. Residual changes in observed output, those unexplained by growth in observed inputs, are called the Solow residual ([Solow](#)) and are imputed to productivity growth by solving

$$g_A(t) \triangleq g_Y(t) - \sum_{i \in \mathbf{I}} \gamma_i g_i(t), \quad (4)$$

where $\hat{\gamma}_i$ and $\hat{g}_i$ denote estimates of output elasticities and growth in factors, respectively. When growth in output, all inputs, and their associated output elasticities are estimated without measurement error, $\mathbf{I} = \mathbf{X}$, growth in the Solow residual is an unbiased estimator of TFP growth in (3). Versions of equation (3) are used by statistical agencies like the Bureau of Labor Statistics in the United States ([Meyer and Harper](#)) and the European Statistical Office in the European Union ([European Commission and Eurostat](#)) to construct measures of productivity growth in national accounts, and in academic applications for comparing productivity across countries and over time ([Feenstra, Inklaar, and Timmer](#)). This approach to measurement typically omits environmental inputs outside of those that produce marketed goods and services where factor income is observed in government accounts.⁴ These estimates of historical TFP growth inform growth predictions that shape the revenue side of government budget forecasts (e.g., 10-year budget forecasts done by the Congressional Budget Office or the President of the United States [Congressional Budget Office; Council of Economic Advisers](#)) used to balance future outlays against receipts.

A critical aspect of the neoclassical production function as a modeling device is that the factor inputs in equation 1 are flows of *services* put towards production (e.g., the number of cycles by a turbine, total hours worked by the labor force, or the amount of crude oil or groundwater extracted). These flows are distinct from the stocks that create them; idle machinery, unemployed persons, and dormant subsurface minerals are not counted as inputs to production. However, in many cases measuring the totality of service flows entering production from a given resource is difficult. For example, the total harvest from a fishery in a given year may account for the harvest as an input in the fishing industry, but it will not capture the service provided by the remaining stock of fish whose reproduction helps replenish the fishery going into the next year ([Hannesson](#)). Total crude oil production will capture the input of extracted petroleum, but will fail to measure how a lower level of reserves reduces underground pressurization and increases future extraction

⁴A unified framework would have to account for non-market services, such as the absorption of man-made pollution, when accounting for the totality of inputs into producing all goods and services humanity consumes.

costs (Anderson, Kellogg, and Salant). In practice, changes in the flows of unmeasured services are often imputed using changes in the capital stocks which produce them.⁵ Growth across all factor inputs can then be proxied by the change in how all stocks of resources—natural, human, or produced—are used over time.

If national accountants were able to calculate the dynamics of all existing natural capital stocks and their output elasticities in production, setting $\mathbf{I}$ to include all observed stocks would always improve their understanding of innovation vis-à-vis any other candidate for $\mathbf{I}$ (see Section 1 of the Supplemental Appendix). Therefore, a goal may be to measure all natural capital stocks and include them all at once (Nordhaus and Kokkelenberg). However, in the interim, limits on institutional capacity for the scope and precision of measurement will likely cause the set of measured inputs $\mathbf{I}$ to differ from $\mathbf{X}$ and cause some error when estimating the growth rates and output elasticities of factor inputs.⁶ Consider a case where a national accountant includes only two inputs, produced capital and labor, when estimating TFP (i.e., in our running example setting $\mathbf{I} = \{K, L\}$ and omitting both forms of natural capital). Conditional on her choice of inputs ($\mathbf{I}$) and her estimates of factor input growth rates and output elasticities ($\hat{\mathbf{g}}, \hat{\boldsymbol{\gamma}}$) she applies, the Solow residual-based estimator of TFP growth can be decomposed as

$$g_A(t)|\mathbf{I}, \hat{\boldsymbol{\gamma}}, \hat{\mathbf{g}} = \underbrace{g_{TFP}(t)}_{\text{True innovation}} + \underbrace{\sum_{x \in \{N_1, N_2\}} \gamma_x g_x(t)}_{\text{Omitted natural capital inputs}} + \underbrace{\sum_{i \in K, L} [\gamma_i g_i(t) - \hat{\gamma}_i(t) \hat{g}_i(t)]}_{\text{Measurement error}}. \quad (5)$$

When natural capital inputs are omitted, (or in the general case whenever $\mathbf{X} \setminus \mathbf{I} \neq \emptyset$) (3) is a biased estimator for TFP growth; the contributions of the omitted natural capital inputs are misattributed to TFP because they are spuriously absorbed by the Solow residual. As a result, output growth driven by (potentially unsustainable) drawdowns of resources may be conflated with (sustainable) technological change. This misattribution can effect three negative outcomes for policymakers irrespective of environmental concerns: 1) an overestimation of the ability for society to substitute away from natural resources (Nordhaus and Tobin); 2) a failure to account for the depletion of natural resources required for replicating historical development paths (Brandt, Schreyer, and Zipperer); and 3) misallocation of resources (Hsieh and Klenow). In turn, a residual in (5) displaying substantial correlation with changes in a given natural capital stock over time is evidence that the natural asset is an omitted input in production. Conversely, a precise null of zero correlation between natural capital stocks and estimated TFP growth is suggestive that the empirical growth accounting equations contain all factor inputs (i.e., $\mathbf{X} \subseteq \mathbf{I}$) and changes in natural capital are unlikely to bias current measures of growth. We now turn to describing a null hypothesis of zero correlation between natural capital and existing measures of TFP in the data.

⁵This imputation is further complicated by the fact that the levels of stocks typically influence market or shadow prices of services entering production.

⁶It is also possible that there are natural capital stocks which produce services necessary for output which that current science is not aware of.

1.1 Hypothesis Testing

Existing measures of productivity in national accounts and research datasets are calculated using Solow residual-based methods that ignore natural capital (Meyer and Harper; Feenstra, Inklaar, and Timmer; European Commission and Eurostat). As shown in the decomposition in (5), these methods lead to three pathways for natural capital to be correlated with measured changes in TFP: (1) correlation between natural capital use and unobservable growth in TFP, (2) the omission of natural capital stocks which should enter (5), and (3) correlation with measurement error when calculating growth rates or output elasticities of included factors. While we cannot distinguish between these three potential mechanisms, we can test whether the Solow residual-based estimator of TFP is uncorrelated with changes in natural capital stocks. If growth in natural capital is correlated with TFP growth, we cannot rule out channel (2). Such a finding implies that improved measurement of natural capital is less likely to be immaterial for measuring TFP.

In an ideal world we would test channel (2)—whether N_i is an omitted variable—in isolation. However, as we only observe g_A jointly as the sum of the three terms in (5), we are restricted to testing whether the correlations between growth in natural capital stocks and *measured* TFP are nonzero. This leads to two potential sources of type one errors where we correctly reject the null hypothesis of no correlation between g_A and g_{N_i} but spuriously attribute it to N_i being omitted. The first two cases correspond to channels (1) and (3); when g_{N_i} is systemically correlated with true innovation, measurement error in the growth accounting equation, or both. Finally, there is a potential source of type two errors: if g_{N_i} is an omitted variable in the growth accounting equation and correlated with either measurement error or TFP growth, it is possible these correlations offset one another. The econometrician may then correctly fail to reject the null hypothesis of no correlation between g_A and g_{N_i} , but spuriously attribute this to natural capital playing no role in production.

Other theoretical barriers exacerbate the challenge of falsifying that estimates of g_A absorb the growth effects of changes in the use of natural capital. First, estimating output elasticities γ_x is nontrivial even for the case of physical capital where factor income can be attributed (Caselli and Feyrer); estimation is more difficult for labor and material inputs and at best a nascent practice for natural capital (Arrow et al.). In short, it is difficult to rule out that measurement error in factor input growth that enter existing estimates of TFP growth is non-classical or reject that it is correlated with growth in natural capital. Second, identifying the growth of natural capital inputs requires delineating what constitutes the flow of natural capital services that enter production from traditional measures of natural capital, such as stocks. For example, the amount of local cooling or precipitation provided by agriculture-adjacent forests may change substantially year-on-year despite de-minimus changes in canopy cover or biomass (Grosset, Papp, and Taylor). Finally, correlations between input growth and the theoretical object g_{TFP} are untestable, as the latter cannot be observed. While measures of some forms of innovation exist (e.g., patent filings and seed varieties Acemoglu et al.; Moscona and Sastry), we cannot rule out underlying correlations between environmental change and the broad measure of technological progress we seek to estimate.

Our estimation strategy below relies on three assumptions in order to be of use in testing whether natural capital inputs are omitted: (1), measurement error in the (weighted) factor input growth terms used to construct estimates of TFP is classical; (2), changes in natural capital stocks are a good proxy for changes in the flow of services they produce; and (3), the rate of true technical change g_{TFP} is conditionally uncorrelated with the changes in environmental inputs to production we consider. These assumptions are strong ones; however, their failure would not be indicative of natural capital being irrelevant for national accountants. When measurement error in non-classical and/or correlated with natural capital (assumption 1 fails), better measurement of natural capital will improve the measurement of factor inputs that currently enter national accounts even if it has no direct value when calculating productivity growth. If changes in existing measures of natural capital stocks are poor proxies for the flows of environmental services essential for production (assumption 2 fails), it is a powerful argument for adjusting the natural capital stocks we measure in order to keep better track of the aspects of the environment which are important for the economy. Finally, if natural capital inputs cause changes in TFP (assumption 3 fails), measurement of natural capital should be a first-order priority for national accountants in order to determine how nations may adjust the use of the environment to accelerate growth. The failure of any of the above assumptions should instead be seen as additional evidence for the importance of measuring natural capital independent of measuring productivity growth.

1.2 Estimating Equation

To test the hypothesis of whether growth in natural capital stocks is associated with growth in measured TFP, we regress measured country-level TFP growth ($g_{A_{j,t}}$) from the Penn World Tables (PWT, [Feenstra, Inklaar, and Timmer](#)) on measured growth in seven country-level natural capital stocks ($g_{N_{n,j,t}}$) taken from the 2024 vintage of the World Bank CWON dataset. Under the second assumption described in Section 1.1 (that growth in natural capital stocks is directly proportional to the growth in the services that they provide) this allows us to identify the estimands γ_i . Equation (6) shows the empirical analog of (5) that can be taken to available data; we estimate (6) on a balanced panel of 117 countries over the 1996-2019 period:⁷

$$g_{A_{j,t}} = \sum_{n \in \mathbf{N}} \gamma_n g_{N_{n,j,t}} + \sum_{z \in \mathbf{Z}} \theta_z g_{z,j,t} + \delta_j + \zeta_t + \varepsilon_{j,t} \quad (6)$$

where j and t index countries and years respectively. The set $\mathbf{N}$ contains growth in the seven natural capital inputs at the country level listed in Table 2 of the Supplemental Appendix and coefficients γ_i govern their relationship with TFP growth. The summation over terms $g_{z,j,t}$ allows for controls for growth in produced capital, employment, the labor share, and human capital at the

⁷The CWON dataset in its raw form contains an unbalanced panel of natural capital stocks in each country covering 1995-2020. We impute missing values as zero changes from the last period where measurements were available leaving us with a balanced panel of growth in natural capital stocks between 1996 and 2020 (see Appendix Section 2.2 and 2.3 for a discussion of the imputation process for missing data along with robustness checks to ensure the results are not an artifact of this process). Data on total factor productivity levels in the PWT version 10.01 only cover years out to 2019, as such our estimating sample covers 1996-2019.

country-by-year level taken directly from the PWT (Feenstra, Inklaar, and Timmer). Terms (δ_j, ζ_t) allow for country and year fixed effects (labeled TWFE in Table 1), and $\varepsilon_{j,t}$ are idiosyncratic error terms that we assume to be conditionally uncorrelated with growth in the natural capital stocks. We note here explicitly that our estimands of interest (the γ_i terms) are not the causal effects of changes in natural capital stocks on TFP growth. The estimation here does not seek to test whether natural capital inputs cause productivity growth or estimate the output elasticities associated with various natural capital stocks. These coefficients identify the (conditional) covariance between growth in natural capital stocks and growth in TFP. Rejecting that these covariances are zero gives us statistical evidence that a given natural capital stock is correlated at least one of the components of the decomposition of TFP growth in (5).

Table 1 shows the estimates of $\hat{\gamma}$ from various specifications of (6), each of which gives how a change in the growth of a given natural capital stock affects expected growth in TFP. To test whether natural capital stocks are omitted variables when accounting for output growth, our estimand of interest is the vector of coefficients on growth in natural capital $\gamma = [\gamma_1, \dots, \gamma_7]'$. The null hypothesis is then:

$$H_0 : \gamma = \mathbf{0} \quad (7)$$

where $\mathbf{0}$ is the zero vector. Failing to reject H_0 suggests that the collection of all natural capital stocks in the CWON plays no role in the production function, and omitting them is unlikely to bias current metrics of TFP. Rejecting the null in (7) as suggestive evidence that natural capital assets enter the production function and bias Solow residual-based estimators of TFP when omitted.

The bottom rows of table 1 show p -values from two Wald tests of this joint null hypothesis across all seven natural capital stocks we consider. The first row constructs test statistics using asymptotic standard errors (clustered at the country level), while the second uses a semi-parametric wild bootstrap approach. In specifications with time and country fixed effects (column 3 onward), we can reject that natural capital has no role in explaining output at traditional significance levels under both tests considered. In addition, almost all point estimates are positive, the expected sign if the natural capital stocks we consider are beneficial inputs to production and covary positively with output.⁸ While the estimated joint effects of changes in natural capital appear precise, the overall residual variation explained by adding natural capital on the margin is limited. Including natural capital increases model R^2 by 0.006 (3 percent over baseline levels) for the benchmark fixed effects models in columns (3) and (4).

Robustness: Partial correlations and individual tests of coefficient significance suggest that land rents are the natural capital inputs most likely to be material for current TFP metrics, aligning with suggestions for improving national accounting given by the authors of the latest edition of the PWT (Feenstra, Inklaar, and Timmer). However, even when we take the extreme measure of excluding urban, agriculture, and timber land from the set of natural capital in order to avoid

⁸The negative coefficient on forest cover may be due to larger growth in forest cover reflecting a declining rate of forest land being converted to structures or agricultural land, which are likely positively associated with TFP growth.

any bias due to land use change affecting the measurement of capital income, the null hypothesis is still rejected (see Section 2.3 of the Supplemental Appendix). Additional tests verifying the robustness of the results for other subsets of independent variables (e.g., excluding natural capital stocks with a high share of missing or imputed values, expanding them to include non-renewable natural capital), choice of metric for TFP (e.g., measures from national statistical agencies and industry-specific industry based measures), and sample period can be found in Section 2.3 of the Supplemental Appendix.

2 The Marginal Inclusion of Natural Capital

Section 1 gives evidence that the omission of natural capital biases existing measures of TFP. We now examine under what conditions including observed changes in natural capital improve estimated productivity growth. How much do these omissions matter for the precision of our existing estimates, and under what conditions will existing estimates be improved through including the limited set of natural capital stocks we measure today? While intuition suggests that including a previously-omitted determinant of growth is expected to improve our estimates of TFP, it is not always the case. Consider the residual based growth estimator in our running four-input example. Suppose an accountant is able to measure growth in output, labor, produced capital, and non-renewables as well as the associated output elasticities without error, but cannot measure growth in the service flows to society from renewable resources over time (g_{N_2}). If the accountant elects to omit non-renewables when calculating TFP growth, substituting this into (5) yields estimand

$$\hat{g}_A(t) \triangleq g_{TFP}(t) + \gamma_1 g_{N_1}(t) + \gamma_2 g_{N_2}(t) \quad (8)$$

If the accountant instead includes non-renewables (g_{N_1}), the estimand for TFP becomes

$$\tilde{g}_A(t) \triangleq g_{TFP}(t) + \gamma_2 g_{N_2}(t) \quad (9)$$

Clearly, it is ambiguous whether (9) is a less biased estimate of g_{TFP} than (8) without making an assumption on the sign of how the use of renewable inputs has changed over time. Section 1 of the Supplemental Appendix shows how the results generalize when multiple inputs may be simultaneously omitted; bias is reduced only when all omitted terms share a sign in expectation. Criteria for improving in-sample metrics of goodness of fit (such as root mean squared error) are even more stringent, as they depend on higher moments for growth in the omitted stocks and the set of included inputs. The effects of including natural capital on the margin, even with perfect measurement of growth and estimation of output elasticities, are only guaranteed to improve estimates of TFP when it leads the estimator to be correctly specified (i.e., $\mathbf{X} \subseteq \mathbf{I}$).

The result that marginal inclusion of natural capital stocks need not improve estimates of TFP growth (or output elasticities) is a central addition to this literature. For instance, existing work by [Brandt, Schreyer, and Zipperer](#) shows that under the assumption that non-renewable capital inputs

are uncorrelated with the use of any other natural capital stocks, TFP growth inclusive of non-renewable natural capital tends to be greater than when it is measured excluding non-renewable natural capital. However, because of the results shown here, it is not possible to know whether these reconstructed series that adjust for non-renewable resources bring estimates closer to true productivity growth over time.

3 Calibrated Estimates of Incorporating Natural Capital on the Margin

Section 2 showed that the effects of marginal additions of inputs to the growth accounting equation, even when perfectly measured, are not guaranteed to improve residual-based TFP estimates. The final section here examines when the addition improves estimation conditional on the behavior of observed growth in natural capital in the data. Here, we examine theoretical conditions under which measuring the growth rate of a natural capital stock used in production and including it on the margin improves estimation of TFP growth. Our laboratory returns to the model economy governed a four-factor production function. Like in the example of Section 2, suppose a national accountant seeks to measure TFP growth in this economy, but measures only growth in the usage of produced capital, labor and non-renewable capital, N_1 . Growth in all of these quantities (along with output) is measured without error. While the national accountant suspects N_1 is important in production, she has no ability to measure the use of renewable resources (N_2) over time. Knowing this, she must choose whether to include growth in societal use of non-renewable resources, N_1 , when taking Equation (5) to the data. The accountant’s criteria for improved TFP measurement is expected in-sample mean squared error for her data, which cover 25 years.⁹

To evaluate how the decision to include natural capital may affect the resulting measurement error in her estimates of TFP, we conduct a thought experiment where each estimator of applied TFP growth is applied to simulated processes for output growth calibrated to historical values across a broad set of countries. To operationalize this process, we follow the World Bank CWON methodology (World Bank) and construct aggregate quantity indices for non-renewable and renewable resource use at the country level and their growth, $g_{N_1,j,t}$ and $g_{N_2,j,t}$, over time (see Section 3 of the Supplemental Appendix for details). We estimate empirical averages and correlations for the historical use of non-renewable capital and renewable capital in 100 countries for the period between 1996 and 2019. We pair these estimates for the average historical growth in natural capital use with averages of measured growth in produced capital, labor, and TFP taken from the Penn World Tables (Feenstra, Inklaar, and Timmer) for the same period.

After calibrating the data generating processes for each country, we measure the effect of introducing non-renewable resources N_1 in the growth accounting equation on the root mean squared error (RMSE) in residual-based estimates of TFP growth, when output elasticities are estimated

⁹We round up from the size of 24 years in our estimating sample.

using ordinary least squares.¹⁰ This, in effect, allows us to compare the accuracy of estimates generated by (8) against (9). We evaluate the expected RMSE at the sample analogs of population moments input growth under standard assumptions for the output elasticities of various factors of production. This gives us a cross section of 100 countries for the effects of including natural capital when calculating TFP growth, which can be conditioned on observed levels and correlations between the use of natural capital over time at the national level.

3.1 Results

Figure 1 plots the results from our calibrated thought experiment. Each cell overlain with an ISO3 code corresponds to the pair of empirical correlations between the growth of non-renewable capital inputs and produced capital inputs (x axis) and the growth of non-renewable and renewable capital inputs (y axis) we observe in the data for that country between 1996 and 2019. For each cell with an ISO code, the outcome is the change in the expected RMSE for estimated TFP growth when non-renewable capital is included on the margin when taking versions of the estimators (8) and (9) to the data.¹¹

$$\Delta\text{RMSE}|\mathbf{g} \triangleq \sqrt{\mathbb{E}[(\hat{g}_A - \mathbb{E}[g_{TFP}])^2|\mathbf{g}]} - \sqrt{\mathbb{E}[(\tilde{g}_A - \mathbb{E}[g_{TFP}])^2|\mathbf{g}]} \quad (10)$$

A positive change in our RMSE metric for a given hypothetical data generating process for growth in factor inputs and TFP $\mathbf{g}$ indicates that including non-renewable resources improves a national accountant’s estimate. The z axis in Figure 1 plots these conditional changes in expected RMSE— $\Delta\text{RMSE}|\mathbf{g}$ —for each country in our calibrated cross section; darker blue indicate where including non-renewables worsens measurement on the margin, while progressively lighter greens show the degree of improvement. The remaining values within the convex hull formed by the cross-section (dashed gray lines) are correlation pairs that do not correspond to the historical behavior of a specific country and are fitted using linear interpolation. The improvement surface then posits the improvement for a hypothetical country when a national accounting includes non-renewable capital (while knowingly omitting renewable natural capital) when calculating TFP conditional on historical correlations in the growth of capital inputs over time.

Figure 1 shows the simulated results for measurement. The most striking result in Figure 1 is that for 93 out of 100 countries, the simulations indicate that including non-renewable resources improves the accuracy of estimated TFP growth. For the seven countries where including non-renewables raises RMSE, the reduction in accuracy is small (less than 0.1 basis point both on average and at the median).¹² In contrast with this, the median and mean improvements in measurement are sizable: 10 and 38 basis points respectively. The improvement is disperse (s.d.

¹⁰See Section 4.3 of the Supplemental Appendix for the full procedure for applying (8) and (9) to our simulated data.

¹¹The full derivation of this process is relegated to Section 4 in the Supplemental Appendix.

¹²The largest RMSE increase occurs for Chile, where RMSE rises by 5 basis points, and is driven by the fact that the correlations between produced capital and natural capital $\rho(g_K, g_{N_1}) = 0.30$ and $\rho(g_K, g_{N_2}) = -0.29$ offset each other in the bias term when TFP growth is estimated excluding natural capital.

71 basis points) and right skewed.¹³ The magnitude of this improvement is a little over 10% of the magnitude of average TFP growth, for which the median values are 98 (33) basis points for countries where TFP growth is positive (negative) on average for the 1996-2019 period.

Figure 2 shows scatter plots illustrating the cross-sectional variation in the relative performance of the estimators against individual country-level characteristics such as trends in resource use and baseline levels of TFP growth. Panels 2a and 2b plot improvements in estimates of TFP growth against various average growth in g_{N_1} and g_{N_2} at the country level. 68 countries exhibit positive average growth in their use of non-renewables and 47 exhibit positive average growth in the use of renewable inputs. Panels 2c and 2d show point estimates of TFP growth based on historical moments from the two estimators against the true data generating process (set as the observed value in the PWT). Panels 2e and 2f shows the improvement from including non-renewable resources against the baseline error in the model omitting natural capital as well as the relative size of this improvement. The median (mean) improvement is 23 (32) percent of baseline error when both natural capital stocks are omitted, and rises to over 60% for over a quarter of the countries in the cross section of hypothetical outcomes. Additional cross sectional analysis examining how the country-level outcomes in our simulations relate to other factors that play a role in measuring productivity (e.g., statistical capacity and the share of GDP stemming from resource rents) can be found in Appendix 4.5.

The resulting improvement for most countries is of a magnitude that is policy-relevant. The median reduction in RMSE across the 100 countries in our sample is 10 basis points (1/100th of a percentage point). This translates to a forecast error of roughly 1 percentage point of TFP growth over a 10-year horizon. Applying this median value to the United States (which has an RMSE reduction of 18 basis points in our simulation) a 1 percentage point reduction in error is three times the annual Congressional Budget Office (CBO) year-on-year forecast error when predicting budget deficits ([Congressional Budget Office](#)). When this forecast error is compounded over ten years (the standard CBO horizon) for current levels of output, the deviation of revenues from projections is roughly equal in size to the aggregate public and private cost of compliance with the Clean Air Act in the United States between 1970 and 1990 in magnitude, one of the most broad environmental policies enacted globally to date ([U.S. Environmental Protection Agency](#)).

Comparison to existing results in the literature: Our estimates of the mean squared error reduction in estimates of annual TFP are similar to those produced by [Brandt, Schreyer, and Zipperer](#) when they consider adjusting existing metrics for a subset of non-renewable resources we consider here. The authors find including non-renewable capital changes point estimates of annual TFP growth by an average of 2.3 basis points for the panel of 21 countries between 1986 and 2008 that they consider. [Freeman, Inklaar, and Diewert](#) use an earlier version of the Changing Wealth of Nations dataset to calculate the bias improvements when only subsurface minerals and fossil fuels matter for output growth and are added to the growth accounting equation on the

¹³The countries with largest improvements (in terms of how much including non-renewables reduces RMSE) are Kenya, Laos, Israel, Belgium, and Tanzania.

margin. Their findings suggest that omitting non-renewable resources can bias TFP levels by up to 50 percent for resource-intensive economies such as those in OPEC or those with large mining sectors. Finally, for context, our estimates for the reduction in estimation error are about two orders of magnitude smaller than the year-on-year revisions of TFP estimates from the United States’ Bureau of Labor Statistics to their estimates for 2024 ([U.S. Bureau of Labor Statistics](#)).

4 Conclusion

This paper presents evidence for how measuring natural capital can aid governments irrespective of environmental or public health goals. When natural capital affects production, omitting it from national accounts biases measures of economic growth. This supports the original assertion by Nordhaus and Tobin in 1972 that leaving things out (“treating as free things that are not really free”) of national accounts “gives the wrong signals for the directions of economic growth.” ([Nordhaus and Tobin](#)).

The existing literature continues highlight how environmental change broadly affects society and welfare. The literature as mostly focused on responding to high level calls to improve welfare measures, e.g., the U.N. Secretary General’s push for measures “Beyond GDP” ([United Nations Statistics Division](#)). The importance of such measures are recognized by economists as well ([Jorgenson](#)). While measures of the environment are indeed likely for measures of national Hicksian income, sustainable development, and national welfare, these are not the only reasons to systematically connect changes in the environment to economic statistics.

Natural capital has first-order consequences for measuring and forecasting economic growth, even for governments that do not see the health, preservation aspects of environmental, or sustainable development outcomes a priority. The reductions in error from improved measurement will influence objectives such as deficit reduction and growth targets that are almost always concerns for governments irrespective of their position on the intrinsic value of the environment to society.

The size of existing errors, when measuring TFP, are economically meaningful even for horizons as short as ten years. Historical growth in inputs and measured productivity between 1996 and 2019 suggest that over half of countries globally may improve their TFP estimates of TFP by 10 percent or more by including non-renewable resources when estimating productivity growth. Even for countries where this improvement is small, measurement of natural capital in the interim will still help with retrospective calculations of TFP as the full set of inputs governing the economy becomes clearer over time. In the case of the United States, for an “America’s Balance Sheet” policy to improve the quality of economic measurement, it is important to have a broad set of natural capital measures and not simply a narrow focus on mineral resources and surrounding industry ([U.S. Department of the Interior](#)). Comprehensive measurement will bolster federal ability to track productivity growth and correctly price natural and produced assets, accounting for the interdependence between both in production.

Finally, the posterity value of measurement undertaken today may be high even for resources

that lack a role in current production processes. Expanding the set of measured stocks is substantially easier when done contemporaneously (c.f. when done retrospectively). [Fenichel](#) argues a critical feature of natural capital is understanding a small number of accounting boundaries, in part to separate resources currently entering production from those that are not. Yet, innovation itself, measured by TFP, can change what is important for innovation. Having measurements in hand can help address this “new inputs” challenge in measuring the economy ([Trajtenberg](#)). As future generations develop new productive technologies that draw on previously-unused environmental services, improved measurement will enable future generations to achieve commensurate progress in informing future fiscal forecasts.

Table 1: TFP Growth vs. Growth in Renewable Natural Capital

Outcome: $\Delta \ln(TFP_{j,t})$	(1)	(2)	(3)	(4)	(5)	(6)
Urban Land	0.301 (0.095)	0.303 (0.098)	0.205 (0.067)	0.215 (0.075)	0.149 (0.062)	0.060 (0.118)
Timber Land	0.164 (0.090)	0.166 (0.075)	0.071 (0.074)	0.075 (0.071)	-0.003 (0.068)	0.026 (0.104)
Agricultural Land	0.027 (0.025)	0.043 (0.025)	0.030 (0.022)	0.036 (0.023)	0.041 (0.021)	0.043 (0.017)
Forest Cover	-0.067 (0.094)	-0.111 (0.091)	-0.005 (0.079)	0.012 (0.083)	0.083 (0.053)	-0.027 (0.076)
Mangroves	0.122 (0.244)	0.067 (0.242)	0.357 (0.299)	0.327 (0.322)	0.318 (0.367)	-0.108 (0.427)
Biomass	0.112 (0.098)	0.026 (0.077)	0.094 (0.108)	0.072 (0.101)	0.047 (0.135)	-0.104 (0.108)
Hydropower	0.011 (0.004)	0.011 (0.004)	0.009 (0.004)	0.009 (0.004)	0.006 (0.003)	0.006 (0.003)
Controls for . . .						
Factor Inputs	No	Yes	No	Yes	Yes	Yes
TWFE	No	No	Yes	Yes	Yes	Yes
Idiosyncratic Trends	No	No	No	No	Yes	No
Lags of Outcome	No	No	No	No	No	Yes
Wald	0.007	0.003	0.007	0.005	0.002	0.039
Bootstrap Wald	0.092	0.063	0.046	0.021	-	0.056
# Observations	2,808	2,808	2,808	2,808	2,808	2,457
# Countries	117	117	117	117	117	117

Note: The outcome variable is the change in the logarithm of the $RTFP^{NA}$ series in version 10.01 of the Penn World Tables ([Feenstra, Inklaar, and Timmer](#)). All independent variables in the table enter the regression after being transformed into log changes. Rows under the “Controls for . . .” heading denote whether covariates controlling for factor inputs (capital services, labor, human capital, and the labor share), country and year fixed effects (TWFE), and idiosyncratic time trends at the country level are included. The last column controls for two lags of the dependent variable $\Delta \ln(TFP_{j,t})$ and is estimated using the Arellano-Bond (1991) dynamic panel estimator. Standard errors clustered at the country level are shown in parentheses. Row “Wald” displays p values from the joint null hypothesis that coefficients on all natural capital growth terms are zero. Row “Bootstrap Wald” shows p -values from wild bootstrap tests of the joint null hypothesis. No results are reported for this test in column (5) due to the idiosyncratic trend terms.

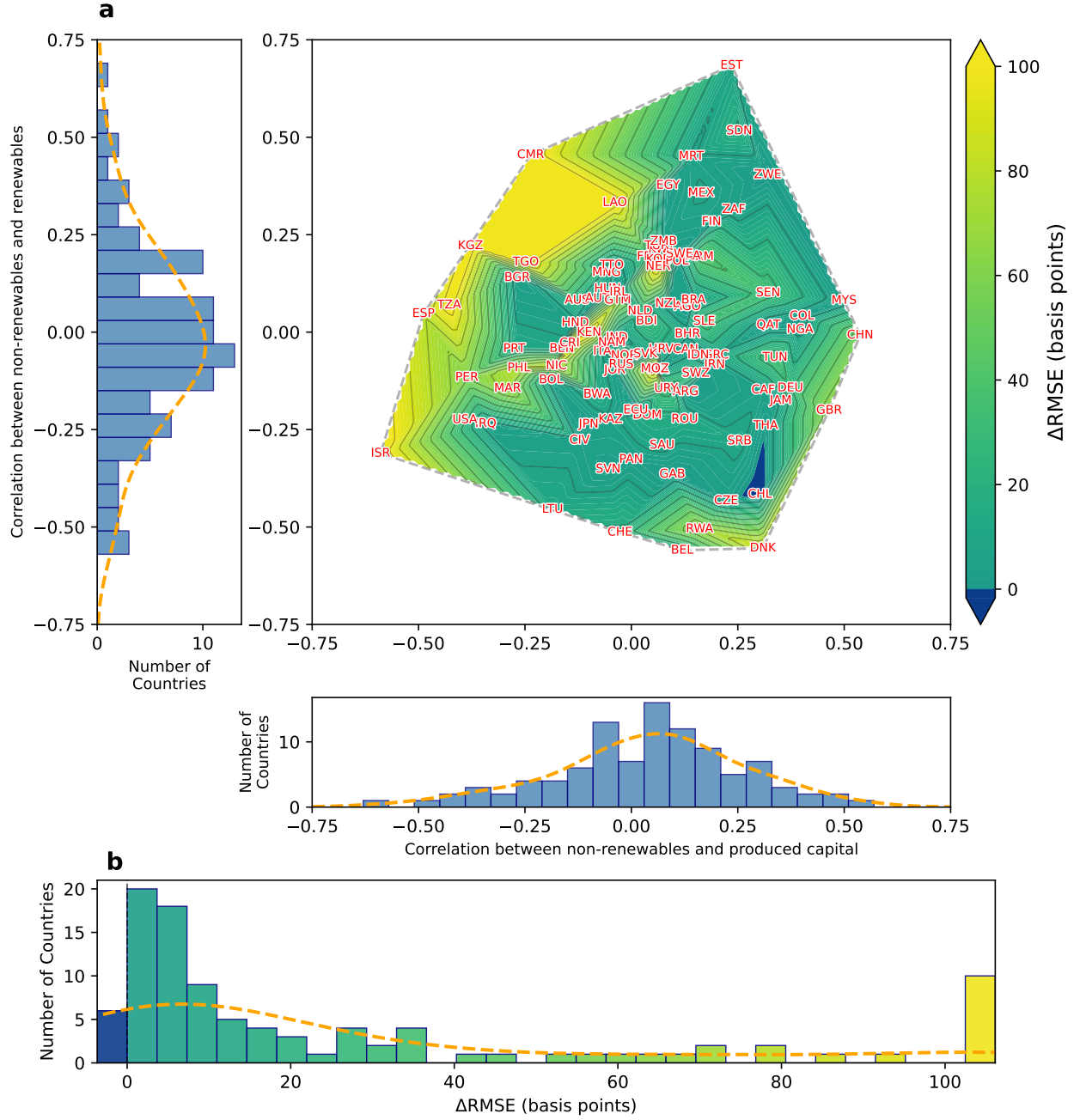


Figure 1: Panel 1a shows a contour plot of $\Delta\text{RMSE}|g$ (measured in basis points) when adding g_{N_1} to the residual-based TFP estimator plotted against observed correlations between growth in non-renewables and produced (renewable) capital at the country level between 1996 and 2019 for the 100 countries we consider. Contour lines denote level sets of the reduction in RMSE. ISO are overlain on the simulated values for countries in our dataset while other points within the convex hull (dashed grey line) are fitted using piecewise linear interpolation after topcoding the z values at the 90th percentile. Histograms on each axis show the empirical distributions of the correlations at the country level. Panel 1b shows a histogram of the reductions in mean squared error shown in Panel 1a with corresponding color coding for each bin.

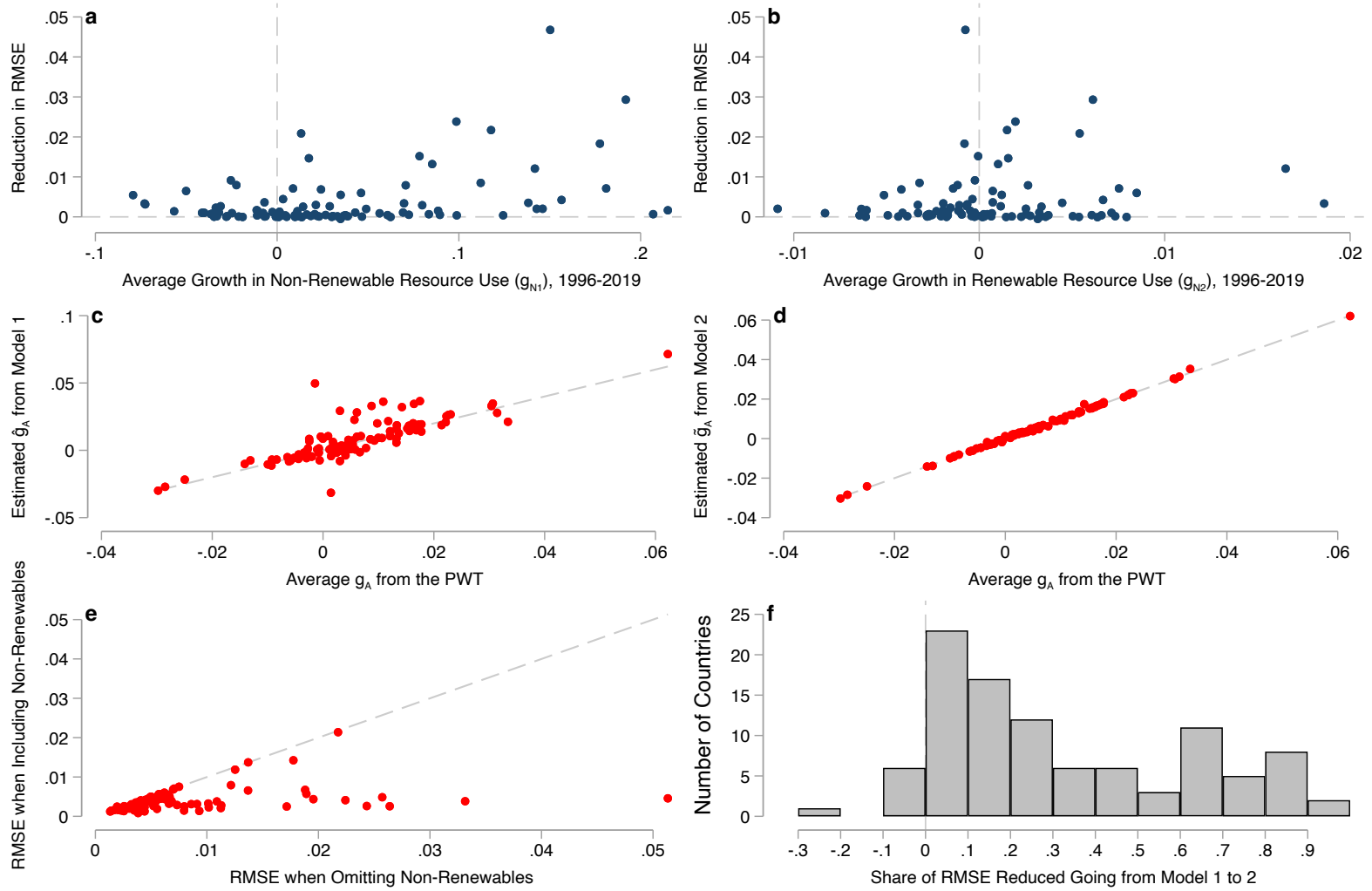


Figure 2: Panels 2a and 2b plot improvements in estimates of TFP growth against average growth in non-renewable and renewable resource use at the country level between 1996 and 2019 for our panel of 100 countries. Panels 2c and 2d show point estimates of TFP growth based on historical moments from the two estimators against the true data generating process. Panels 2e and 2f shows the improvement from including non-renewable resources against the baseline error in the model omitting natural capital as well as the relative size of this improvement. Dashed lines in panels 2a, 2b, and 2f are plotted at $x = 0$ and $y = 0$. Dashed lines in panels 2c-2e show the 45 degree line.

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